



Defecation — MCQ Practice

Physiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A-E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. A 26-year-old opens his bowels three or four times a week, has no pain, and feels well. Based on the lecture, this pattern is best regarded as: EASY

- A. Evidence of a defective parasympathetic defecation reflex
- B. A sign of peritoneointestinal reflex activity
- C. Constipation caused by decreased intestinal motility
- D. Normal, since the range runs from 2–3 per day to 3–4 per week
- E. Diarrhoea caused by decreased absorption

2. A man is sitting quietly when a mass movement forces feces into his rectum. What happens immediately? EASY

- A. Only a voluntary contraction of the external sphincter occurs, with no urge
- B. He feels the desire to defecate, with reflex rectal contraction and relaxation of the anal sphincters
- C. Peristaltic waves reverse toward the sigmoid colon
- D. Reflex rectal relaxation with increased tone of both anal sphincters, delaying the urge
- E. The abdominal muscles contract without any rectal response

3. During anal surgery a surgeon notes a thickened ring of circular smooth muscle just inside the anal canal. Which structure is this? EASY

- A. Rectal valve
- B. Internal anal sphincter
- C. Pudendal nerve
- D. External anal sphincter
- E. Anal column

4. A 22-year-old develops loose stools shortly after drinking milk. Which cause of excessive motility named in the lecture fits best? EASY

- A. Irritation of the peritoneum
- B. Habitual delay of defecation
- C. Inadequate fluid intake
- D. Lactose intolerance
- E. Insufficient dietary fibre

5. After pelvic trauma, a patient has bilateral pudendal nerve injury. Which function is most directly lost? MEDIUM

- A. Voluntary constriction of the external anal sphincter
- B. Afferent signalling from stretch receptors in the rectal wall
- C. Relaxation of the internal sphincter by myenteric inhibitory signals
- D. Tonic constriction of the internal anal sphincter
- E. Peristaltic waves in the descending colon and sigmoid

6. An investigator distends the rectal wall in a preparation containing only the rectum's own nerve network. Through which pathway do afferent signals spread to start peristaltic waves in the descending colon, sigmoid and rectum? MEDIUM

- A. Skeletal motor nerves to the abdominal wall
- B. The pudendal nerve
- C. The myenteric plexus
- D. The pelvic nerves via the sacral cord
- E. The cerebral cortex

7. After a mass movement, a healthy adult empties the large bowel from the splenic flexure to the anus in one powerful effort. Which mechanism provides that strength?

MEDIUM

- A. The intrinsic myenteric reflex acting alone
- B. The peritoneointestinal reflex
- C. The parasympathetic defecation reflex via the sacral cord and pelvic nerves
- D. The renointestinal reflex
- E. Voluntary abdominal contraction acting alone

8. As a healthy adult feels the urge, defecation signals entering the spinal cord trigger a deep breath, closure of the glottis and contraction of the abdominal wall muscles. What is this sequence for?

MEDIUM

- A. Raising intra-abdominal pressure to force colonic contents downward
- B. Constricting the external sphincter to delay defecation
- C. Directly relaxing the internal anal sphincter
- D. Suppressing the urge to defecate
- E. Converting the parasympathetic reflex into an intrinsic reflex

9. A 38-year-old with acute kidney irritation has markedly reduced bowel activity. Which reflex best explains this?

MEDIUM

- A. Peritoneointestinal reflex
- B. Vesicointestinal reflex
- C. Intrinsic myenteric defecation reflex
- D. Parasympathetic defecation reflex
- E. Renointestinal reflex

10. A sedentary 45-year-old with low fibre and low fluid intake passes infrequent, dry, hard stools. Which mechanism best explains the dryness?

MEDIUM

- A. Loss of pudendal control of the external sphincter
- B. Rapid transit of chyme through the small intestine
- C. Feces stay in the colon for a prolonged period, so excessive water is absorbed
- D. Excessive motility leaves too little time for absorption
- E. Overactivity of the peritoneointestinal reflex

11. A 40-year-old with constipation asks for a long-term solution. Which approach does the lecture describe as safer than laxatives?

MEDIUM

- A. Daily laxatives, because they are not habit-forming
- B. Delaying defecation until it is convenient
- C. Relying on voluntary abdominal straining alone
- D. Reducing fibre to slow colonic motility
- E. Adding fibre, exercising more and drinking more fluid

12. A 25-year-old has a spinal cord transection and cannot consciously control his external anal sphincter. How is his lower bowel emptied?

MEDIUM

- A. Automatically, by the defecation reflexes, without voluntary control
- B. Only by deliberate abdominal contraction
- C. Not at all, because the intrinsic reflex needs cortical input
- D. By continuous dribbling, because the internal sphincter loses tone
- E. Only through the peritoneointestinal reflex

13. During a meeting, a woman's rectum fills after a mass movement and the peristaltic wave reaches the anus with relaxation of the internal sphincter, but she keeps her external sphincter voluntarily constricted. What happens?

MEDIUM

- A. The pudendal nerve abolishes the peristaltic wave
- B. The internal sphincter re-contracts and the reflex is cancelled permanently
- C. Defecation does not occur, because the external sphincter has not relaxed
- D. Defecation occurs, because the parasympathetic reflex forces the external sphincter open
- E. Defecation occurs, because internal sphincter relaxation alone is sufficient

14. A colorectal surgeon describes a weak functional sphincter about 20 cm from the anus that helps keep the rectum empty. Where is it located?

MEDIUM

- A. At the junction of the descending and sigmoid colon
- B. At the level of the external anal sphincter
- C. At the splenic flexure of the colon
- D. At the junction between the sigmoid colon and the rectum
- E. Within the anal columns

15. A patient has a lesion of the parasympathetic pelvic nerves; the enteric nervous system of the rectal wall and the pathways above it are intact. Which outcome best fits?

HARD

- A. Defecation relies on the intrinsic reflex, which is relatively weak alone, so emptying is less effective
- B. Defecation is impossible, because the intrinsic reflex needs the pelvic nerves
- C. Voluntary control of the external sphincter is lost, as in newborns
- D. Defecation is unchanged, because the parasympathetic reflex adds nothing to the intrinsic reflex
- E. Automatic, uncontrolled emptying occurs whenever the rectum fills

16. Two students compare the defecation reflexes. Which statement is correct?

HARD

- A. Both reflexes act through the pudendal nerve to relax the external anal sphincter
- B. The intrinsic reflex is the powerful one, and the parasympathetic reflex only adds voluntary control
- C. The parasympathetic reflex contracts the external sphincter, while the intrinsic reflex relaxes the internal sphincter
- D. Both reflexes depend on the sacral cord and differ mainly in how quickly they act on the colon
- E. The intrinsic reflex is enteric and weak alone; the parasympathetic reflex is sacral and powerful, emptying from the splenic flexure to the anus

17. A patient receives a complete pudendal nerve block for a perineal procedure. What is expected of continence at rest?

HARD

- A. Internal sphincter tone is lost while the external sphincter stays constricted
- B. External sphincter tone is lost, but tonic constriction of the internal sphincter persists and still limits dribbling
- C. Tonic constriction of both sphincters is lost, so dribbling is inevitable
- D. No change, because both sphincters are maintained entirely by the myenteric plexus
- E. The internal sphincter relaxes continuously because of parasympathetic drive

18. Which sequence correctly describes the parasympathetic defecation reflex?

HARD

- A. Abdominal muscle contraction → glottis closure → sacral cord → internal sphincter relaxation
- B. Rectal nerve endings stimulated → cerebral cortex → pudendal nerve → contraction of the external sphincter
- C. Rectal distension → myenteric plexus only → weak peristalsis, with no sacral cord involvement
- D. Rectal nerve endings stimulated → sacral cord → pelvic-nerve parasympathetic fibres → stronger peristalsis and internal sphincter relaxation
- E. Rectal nerve endings stimulated → sacral cord → pudendal nerve → colonic peristalsis

19. One patient has bladder irritation and another has acute peritonitis; both have reduced bowel activity. Which statement is correct?

HARD

- A. Peritonitis acts through the vesicointestinal reflex and bladder irritation through the peritoneointestinal reflex
- B. Bladder irritation inhibits gut activity via the vesicointestinal reflex; peritoneal irritation inhibits excitatory enteric nerves and can cause paralysis
- C. Both reflexes increase intestinal activity and cause diarrhoea
- D. Bladder irritation acts through the renointestinal reflex, and peritonitis through the intrinsic myenteric reflex
- E. Both act through the parasympathetic defecation reflex and the pelvic nerves, converting weak peristalsis to strong

20. Patient X passes frequent large-volume watery stools; patient Y passes infrequent dry, hard stools. Which pairing of motility and water handling is correct?

HARD

- A. X: decreased motility with excessive absorption; Y: increased motility with too little absorption
- B. X: decreased motility with too little absorption; Y: increased motility with excessive absorption
- C. Both have increased motility, but X absorbs less water than Y
- D. X: increased motility with excessive absorption; Y: decreased motility with too little absorption
- E. X: increased motility with too little time for absorption; Y: decreased motility with excessive water absorption

✓ Answer key

Q	Answer	Level	Why (lecture slide)
1	D	EASY	Normal bowel activity ranges from 2–3 movements per day to 3–4 per week. [Slide 12]
2	B	EASY	A mass movement into the rectum brings immediate desire, reflex rectal contraction and relaxation of the anal sphincters. [Slide 4]
3	B	EASY	The internal sphincter is a several-cm thickening of circular smooth muscle inside the anus. [Slide 5]
4	D	EASY	Excessive motility may be caused by lactose intolerance, stress and microbes irritating the GI mucosa. [Slide 14]
5	A	MEDIUM	The external sphincter is striated, voluntary and controlled by the pudendal nerve (somatic). [Slide 5]
6	C	MEDIUM	Intrinsic reflex: rectal distension → afferent signals through the myenteric plexus → peristaltic waves. [Slide 6]
7	C	MEDIUM	Parasympathetic signals turn the weak intrinsic reflex into a powerful one that empties from the splenic flexure to the anus. [Slide 9]
8	A	MEDIUM	Abdominal wall contraction forces the fecal contents of the colon downward; abdominal pressure rises. [Slides 10, 11]
9	E	MEDIUM	The renointestinal reflex inhibits intestinal activity as a result of kidney irritation. [Slide 13]
10	C	MEDIUM	Constipation: prolonged colonic residence → excessive water absorption → dry, hard feces. [Slide 15]
11	E	MEDIUM	Laxatives are habit-forming; more fibre, exercise and fluid are safer ways of controlling constipation. [Slide 15]
12	A	MEDIUM	In newborns and in cord transection the defecation reflexes cause automatic emptying (no conscious sphincter control). [Slide 12]
13	C	MEDIUM	Defecation occurs if the internal sphincter is relaxed AND the external sphincter is also voluntarily relaxed. [Slide 7]
14	D	MEDIUM	A weak functional sphincter ~20 cm from the anus at the sigmoid-rectum junction, plus sharp angulation. [Slide 3]
15	A	HARD	The intrinsic myenteric reflex is relatively weak alone; parasympathetic signals via the pelvic nerves make it powerful. [Slides 7–9]
16	E	HARD	Intrinsic: rectal enteric nervous system, relatively weak; parasympathetic: sacral segments + pelvic nerves, powerful. [Slides 6–9]
17	B	HARD	Both sphincters give tonic constriction; only the external is controlled by the pudendal nerve, so internal sphincter tone remains. [Slide 5]
18	D	HARD	Rectal nerve endings → spinal cord → back via parasympathetic fibres in the pelvic nerves → intensified peristalsis + internal sphincter relaxation. [Slides 8, 9]
19	B	HARD	Peritoneointestinal: peritoneal irritation → inhibits excitatory enteric nerves → paralysis; vesicointestinal: bladder irritation → inhibits activity. [Slide 13]
20	E	HARD	Diarrhoea = ↑ motility + ↓ absorption; constipation = ↓ motility, prolonged residence and excess water absorption. [Slides 14, 15]